Who Gets Hacked? The Real Drivers of Cyber Conflict

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Abstract

State-sponsored cyberattacks have now become a central feature of modern geopolitical competition yet we know surprisingly little about what factors make a country a target. This paper asks whether regime type predicts how frequently a country is targeted by state-sponsored cyberattacks. Democracies should face higher targeting rates due to their open information environments, high-value digital infrastructure, and geopolitical alignment against major cyber powers. I test this using OLS regression with regional fixed effects across 2,261 country-year observations drawn from the CFR Cyber Operations Tracker and V-Dem dataset ranging from 2005–2022. Internet penetration, and not regime type, emerges as the dominant predictor, while regime type becomes a significant negative predictor once regional variation is controlled for. These findings suggest that internet connectivity, and not the political system, drives targeting, with implications for how states think about cyber vulnerability and digital infrastructure investment. This paper follows Option 2, testing a hypothesis through regression analysis with theoretically motivated control variables such as economy, attack surface and reporting bias.

I. Introduction

State sponsored cyberattacks have become a central part of modern geopolitical competition, but there is still a lot of disagreement about what makes a country a target. A common assumption is that democracies carry more of the burden. They tend to have open information systems, highly connected digital infrastructure, and they are often on the opposite side of major geopolitical divides from countries like Russia, China, and North Korea. At first glance, that argument seems convincing. But when you actually look at the data, the pattern is not that clean. Some of the countries that are targeted most often are not wealthy Western democracies. Instead, they are countries where internet access is growing quickly but cybersecurity capacity has not kept up. This raises a straightforward question that has not really been answered with systematic evidence: does regime type actually shape how often a country is targeted by state sponsored cyberattacks?

The literature offers two main ways of thinking about this. One view is that democracies are more exposed because of their openness, transparency, and the value of the data they hold, which makes them appealing targets for both espionage and disruption. The other view pushes back on that and argues that what looks like a democracy effect is really about deeper structural factors like wealth, connectivity, and geopolitical position. From that perspective, regime type is standing in for other variables that are doing the real work. Once those are taken into account, the relationship between democracy and targeting should weaken or disappear. What the previous studies have not done, however, is test the targeting side systematically. Existing cross-national studies have modeled who launches cyberattacks and not who receives them. No study has used decades of data and directly tested whether regime type predicts targeting frequency while simultaneously controlling for connectivity, wealth, and regional clustering. That is the gap this paper fills.

This paper addresses that question using country-year data from the CFR Cyber Operations Tracker and the V Dem dataset covering the period from 2005 to 2022. One concern with using such a long time span is whether cyberattacks from the mid 2000s are really comparable to those in the 2020s. The internet has changed a lot, and so have the types of attacks. Earlier incidents were often limited to things like website defacement or basic espionage. More recent operations include supply chain compromises, ransomware attacks on critical infrastructure, and interference in elections. However, the CFR dataset records incidents at a consistent level, focusing on whether a country was targeted in a given year, rather than how severe the attack was. That makes the data usable over time, although it is still something to keep in mind when interpreting the results. The findings show that internet penetration, not regime type, is the strongest predictor of how often countries are targeted. Once regional differences are taken into account, more democratic countries are actually targeted less often than autocracies.

II. Background and Literature

The study of state sponsored cyberattacks is still relatively new, and most of the literature has focused on the offensive side of the issue. Researchers have looked at which states carry out cyberattacks, what kinds of capabilities they develop, and what political or military conditions make these operations more likely. Far less attention has been given to the defensive side, especially what explains why some countries are targeted more often than others. This paper is aimed at addressing that gap.

Within the offense oriented literature, there are two main ways of explaining state sponsored cyber behavior. The first emphasizes capability and opportunity. From this perspective, cyberattacks are driven by strategic considerations, where states choose targets based on access, value, and expected payoff. In a study by Hunter, Albert, and Garrett [1] using data from the CFR Cyber Operations Tracker, find that political, economic, and military factors together help explain when states initiate cyber operations. Their results suggest that targeting is largely strategic rather than driven by ideology. Similarly, Hathcock [2] using a situational crime prevention approach across 246 incidents, shows that attacks are heavily concentrated on government and military targets, reinforcing the idea that valuable and accessible systems matter more than regime type alone.

The second perspective focuses on regime type and democratic vulnerability. Building on arguments from the democratic peace literature, this view suggests that authoritarian states are more likely to target democracies. Democracies are seen as more exposed due to open information environments, decentralized institutions, and heavy reliance on digital systems. Nakayama [3] highlights something else here, arguing that the same openness that increases vulnerability also constrains how democracies can respond without undermining their own norms. In this view, democracies should be targeted more frequently, which is a central expectation evaluated in this paper.

More recent work complicates both of these perspectives. Macdonald and Schneider [4] find that democracies rarely target one another, which is consistent with a cyber version of the democratic peace theory [5]. This suggests that regime type does matter in shaping cyber conflict, but not necessarily in the way vulnerability based theories assume. At the same time, research on cybersecurity capacity and regime characteristics points to more complex relationships. A recent study in Democratization [6] using the V Dem Liberal Democracy Index and multiple measures of cybersecurity performance finds non linear effects, suggesting that the relationship between regime type and cyber outcomes is not straightforward.

This paper contributes to these debates by shifting the focus from attackers to targets and by using a longer time window than most existing studies. Rather than asking who launches cyberattacks, it examines who is targeted and why. Using 17 years of cross national incident data, I test whether the democratic vulnerability hypothesis holds under systematic empirical analysis. The results suggest that the capability and opportunity provides a better explanation of targeting patterns, with internet penetration emerging as a stronger predictor than regime type.

III. Theory and Hypothesis

The conventional view in cybersecurity is pretty straightforward which is that democracies tend to be easier and more attractive targets. There are two main reasons for this, and they reinforce each other.

First, democracies tend to be more exposed from a technical and institutional standpoint. They have open information environments, decentralized government systems, and large private sectors that hold valuable data and intellectual property. All of this creates more entry points for attackers and more to gain if an intrusion is successful. You can see this in real-world cases, like Russian interference in Western elections, Chinese campaigns targeting defense contractors, and North Korean attacks on financial institutions. These aren't random, instead they take advantage of the kinds of systems democracies tend to have.

Second, democracies are often on the opposite side of the geopolitical divide from the countries most associated with state-sponsored cyber activity. Being part of U.S.-aligned alliances or partnerships makes them more likely targets, regardless of their internal vulnerabilities. On the flip side, countries that are politically aligned with major cyber powers are less likely to be targeted by them. In that sense, regime type isn't just about internal structure, it also reflects where a country sits politically.

Put together, this leads to a clear expectation. If democracies are both more exposed and more likely to be chosen as targets, then we should see them experiencing more state-sponsored cyberattacks. Thus I hypothesize:

> **H1:** Democracies are targeted by state-sponsored cyberattacks at higher rates than autocracies.

This is a testable claim. It would not hold up if there's no meaningful relationship between a country's level of democracy and how often it is targeted, or if the relationship goes in the opposite direction once other factors are taken into account. To evaluate it properly, it's important to control for things like economic development, internet usage, and regional differences, since democracies and autocracies differ in more ways than just their political systems.

IV. Data and Methods

To test whether regime type predicts how often a country is targeted by state sponsored cyberattacks, I built a country-year dataset covering the period from 2005 to 2022. The unit of analysis is country-year, so each observation corresponds to one country in a specific year. Countries that appear across the full time range contribute up to 18 observations, which explains why the dataset includes 2,482 observations in the base model even though the number of unique countries is much smaller. I evaluate the hypothesis using OLS regression across two model specifications. OLS is used here primarily for its simplicity and ease of interpretation. The coefficients can be read directly as changes in the expected number of cyber incidents associated with a one unit change in each predictor, which makes the results more intuitive. While the dependent variable is a count, the distribution is not extremely skewed and the sample size is fairly large, so OLS still provides a reasonable approximation. In addition, OLS makes it straightforward to include fixed effects and compare results across different models. Alternative models designed for count data, such as Poisson or negative binomial regression, could also be used, but OLS offers a clear baseline for assessing the relationship. The central question is whether the liberal democracy index is positively and significantly associated with cyberattack frequency once other relevant factors are taken into account.

A. Dependent Variable

The dependent variable is cyberattack targeting frequency, which is measured as the number of state sponsored cyber incidents in which a country is identified as a target in a given year. The data comes from the Council on Foreign Relations Cyber Operations Tracker [7], which is one of the most widely used public datasets on state sponsored cyber activity. The tracker records incidents by attributing state actor, target country, year, and type of operation, drawing information mainly on open source intelligence and news reports. A key limitation is that the dataset relies heavily on Western reporting, which means incidents involving non Western countries are likely undercounted. This introduces a reporting bias that cannot be fully corrected, and I return to this issue in the limitations section. Even with this concern, the CFR tracker remains the most practical and widely accepted source for cross national studies of cyber conflict over this time period.

B. Independent Variable

The main independent variable is regime type, measured using the V Dem Liberal Democracy Index [8]. This index ranges from 0 to 1, with higher values indicating more democratic systems. It combines several dimensions, including electoral competition, civil liberties, judicial independence, and constraints on executive power. This makes it a more detailed measure. V Dem is widely used in comparative politics and provides coverage for nearly all countries in the dataset, which makes it a strong fit for this type of analysis.

C. Control Variables

I include three control variables, each based on the idea that regime type may be linked to other country characteristics that also affect cyberattack targeting.

The first control is economic development, measured as the log of GDP per capita in constant 2015 USD using World Bank data [9]. Wealthier countries may attract more attacks because they have more valuable assets, including intellectual property, financial systems, and sensitive government information. Taking the log helps account for the uneven distribution of income across countries.

The second control is internet penetration, measured as the share of the population using the internet. This is the most important control in the model from a theoretical standpoint. Democracies tend to have higher levels of internet use and availability, so any observed effect of regime type could actually reflect differences in connectivity. Including this variable allows me to separate these effects and test whether regime type still matters once the size of a country's digital exposure is considered.

The third control is government effectiveness, measured using the World Bank Worldwide Governance Indicators score, which ranges from -2.5 to +2.5. I include this variable mainly as a proxy to capture differences in detection and reporting capacity. Countries with more effective governments are more likely to identify, attribute, and report cyber incidents, which means they may appear more often in the dataset regardless of actual targeting levels. This helps address reporting bias, though it does not fully resolve it. Note that that 221 observations are lost between the base model and the full model due to missing data on government effectiveness, reducing the sample size from 2,482 to 2,261 observations.

I also include regional fixed effects in the full model using the V Dem regional classification, which groups countries into ten geopolitical regions. This matters because democracies tend to cluster geographically, especially in regions like Western Europe and North America that also experience high levels of cyber activity. Without accounting for region, the model could incorrectly attribute these patterns to regime type. Regional fixed effects address this by comparing countries within the same region, which removes differences across regions and focuses on variation within them.

D. Models

I ran two regression models. Model 1 is a base OLS specification that includes the liberal democracy index, log GDP per capita, and internet penetration, without regional controls and government effectiveness. Model 2 adds government effectiveness and regional fixed effects.

To include regional fixed effects, I convert the `e_regionpol` variable from the V Dem dataset into a factor in R. This creates a set of dummy variables for regions numbered from 1 to 10. One region is omitted and serves as the reference category. The model then estimates the relationship between regime type and targeting frequency while holding regional differences constant, so the coefficients reflect comparisons within regions rather than across them.

This study does not claim a causal relationship. The data is observational, and there are several challenges, including the risk of omitted variables such as rivalry or alliance ties, as well as the possibility of reverse causality. What the analysis does provide, however, is a consistent picture of how cyberattack targeting is associated with country characteristics over time. This kind of descriptive evidence is still valuable, especially in a field that has focused more on who conducts cyber operations than on who is targeted. The panel structure of the data is an important strength. Since it tracks the same countries over 18 years, it captures patterns that would be easy to miss in a single cross section. Using two model specifications also helps show how adding regional controls changes the results, which makes the comparison itself useful.

V. Findings, Results, and Analysis

A. Geographical Distribution

The first visualization in this project is a choropleth map that shows the total number of state sponsored cyberattacks by country from 2005 to 2022 (Figure 1). Starting with a map makes sense because the first question most readers have is simple: who is actually being targeted? Before getting into models or theory, it helps to see the basic geographic pattern. The map was created in R using incident counts from the CFR Cyber Operations Tracker, aggregated over the full time period. Countries are grouped into six categories, ranging from zero incidents to a maximum of 131 in the United States.

The overall pattern is fairly clear. Most activity is concentrated in North America, Europe, and parts of Asia, while regions such as Sub Saharan Africa, Latin America, and Central Asia show very little recorded activity. This pattern is not random. Countries that appear darkest tend to be more connected, more geopolitically relevant, and more likely to be covered in Western reporting. All of these factors are mixed together, and the map alone cannot separate them. This is exactly why a regression approach is needed.

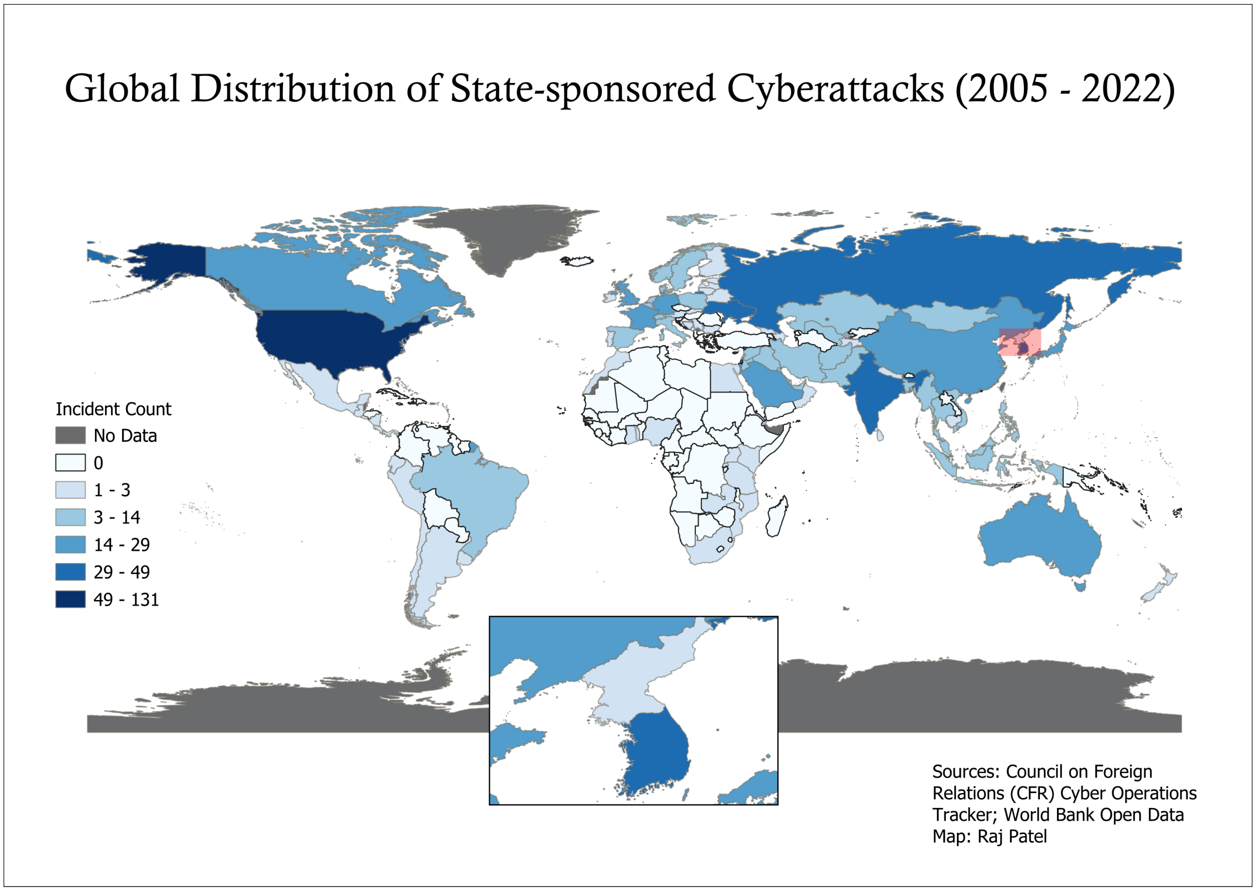

Fig. 1: Global Distribution of State-Sponsored Cyberattacks (2005–2022). Gray indicates no data. Inset shows South and North Korea

There are also some noticeable regional differences. South and Southeast Asia, especially countries like India and South Korea, show relatively high incident counts even though they do not always fit the profile of wealthy Western democracies. This hints that connectivity and geopolitical positioning may matter more than regime type alone. Western Europe shows steady levels of targeting, while the Middle East looks more uneven, with some countries seeing substantial activity and others very little.

One of the more interesting details is the inset map showing South Korea and North Korea. South Korea falls into one of the higher categories, with between 14 and 29 incidents, while North Korea shows almost none. At first glance, this might seem to support the idea that democracies are more vulnerable. But that explanation does not hold up very well, and the North Korean case actually highlights a problem with the data itself. North Korea is widely known to be a target of cyber operations from countries like South Korea and the United States, yet almost none of those incidents appear in the dataset. This is a direct reflection of the reporting bias built into the CFR Cyber Operations Tracker, which relies heavily on Western open-source intelligence and is more likely to capture incidents involving Western-facing targets than those involving closed, non-Western states. To address this, the regression models include government effectiveness as a proxy for detection and reporting capacity, on the assumption that countries with stronger institutions are more likely to identify and publicize incidents. However, as the North Korean case shows, this control can only go so far.

Finally, the gray areas on the map indicate missing data rather than zero incidents. These are countries where the CFR tracker has no recorded cases or where key variables are unavailable. They are excluded from the regression sample, which is important to keep in mind since the results apply only to countries with sufficient data, and those tend to be more visible in global reporting.

B. *Regional Variation in Cyberattack Frequency*

The map shows that cyberattacks are unevenly distributed, but it does not tell us how large those differences are. To make this clearer, Figure 2 presents the average number of attacks by region, along with error bars that show one standard error above and below the mean. A bar chart works well here because it allows for direct comparison while also showing uncertainty.

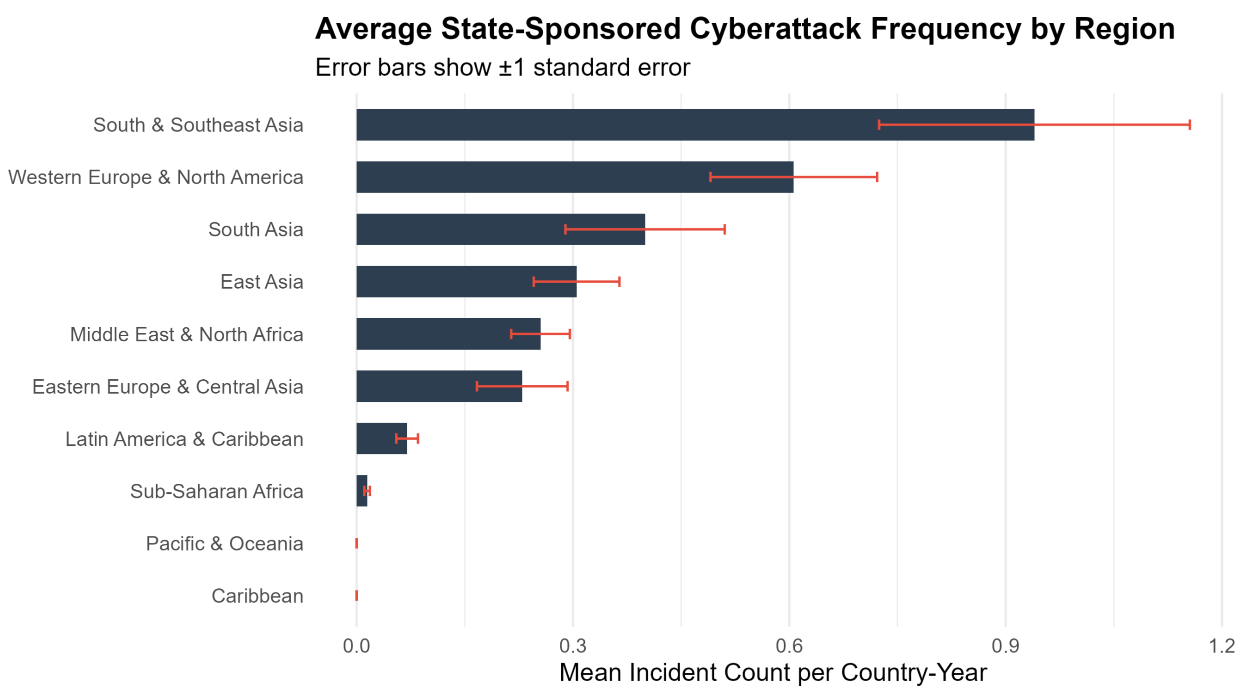

Fig. 2: Average State-Sponsored Cyberattack Frequency by Region (2005–2022). Error bars show ± 1 standard error.

The differences across regions are noticeable. South and Southeast Asia has the highest average, just under one incident per country year, with an error range that reaches close to 1.2. This reflects a big variation within the region, driven by countries like South Korea and India. Western Europe and North America follow at around 0.6 incidents per country year. Given how much attention Western countries receive in discussions of cyber threats, it is kind of surprising that they are not at the top.

Other regions fall in the middle. South Asia, East Asia, the Middle East and North Africa, and Eastern Europe and Central Asia all cluster between about 0.25 and 0.35 incidents per country year. The overlap in their error bars suggests these differences are not too large. At the lower end, Latin America, Sub Saharan Africa, the Pacific and Oceania, and the Caribbean all have averages close to zero.

The gap between the highest and lowest regions is large. If this is not accounted for, regional patterns can be mistaken for country level effects. For example, democracies are heavily concentrated in Western Europe and North America. Without controlling for region, it can look like democracy itself increases risk, when geography is doing much of the work. The bar chart makes this issue clear and explains why regional fixed effects matter and should be included in the regression models.

C. *Cyberattack Trends Over Time by Regime Type*

The regional comparison shows where attacks are concentrated. The next step is to look at how they change over time and whether patterns differ by regime type. Figure 3 plots total incidents by year for democracies, autocracies, and anocracies. A line chart is useful here because it makes it easier to see trends over time. Although the chart includes years beyond 2022, the analysis focuses on 2005 to 2022 to stay consistent with the data used in the models.

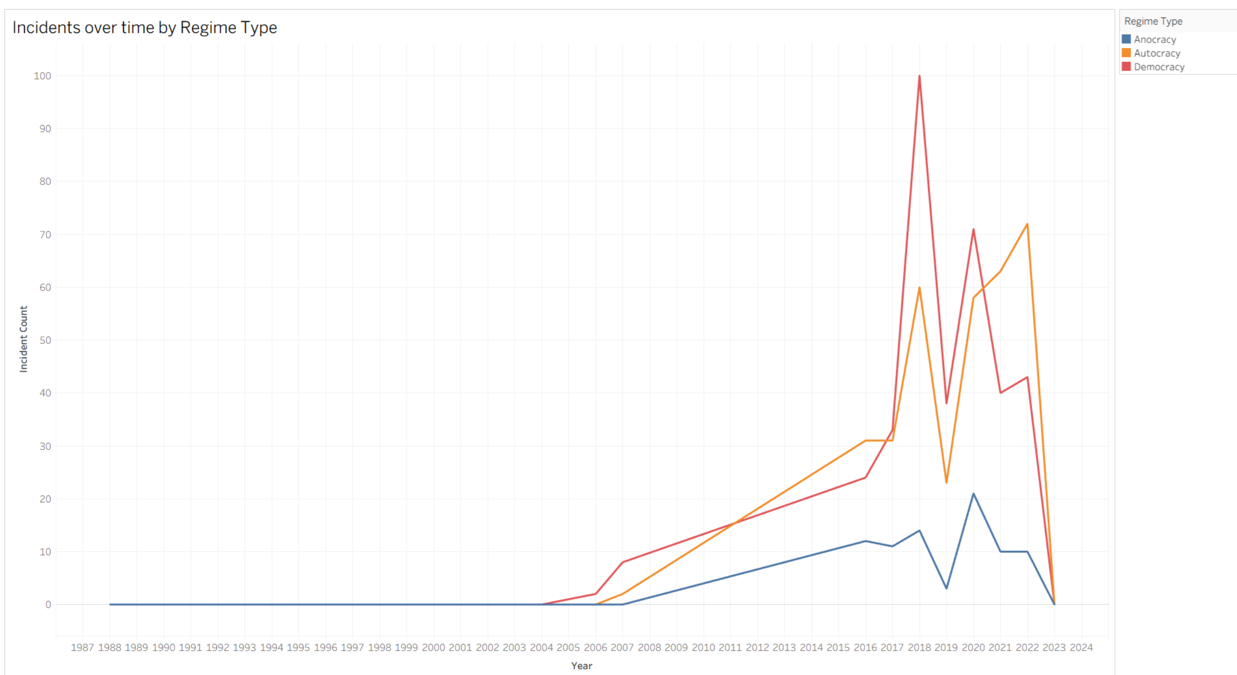

Fig. 3: State-Sponsored Cyberattack Incidents Over Time by Regime Type. Analysis restricted to 2005–2022 to match CFR Cyber Operations Tracker coverage. Regime classification based on V-Dem Liberal Democracy Index.

There is very little activity before 2005. Through the late 1990s and early 2000s, all three lines stay close to zero. This reflects the fact that cyber operations were not yet widely used by states. The internet was not deeply integrated into government or economic systems, so there was less incentive to carry out attacks at scale. This also raises a comparability issue, since incidents in 2005 and 2022 are coded the same way even though the underlying technology is very different.

After 2005, activity increases across all regime types. Democracies and autocracies begin to separate from anocracies fairly quickly. Around 2012, the upward trend becomes more pronounced. This period aligns with several well known developments, including cyber operations linked to conflicts in the Middle East and increased activity tied to Russia before the annexation of Crimea. These cases suggest that cyber operations were becoming a more regular part of state behavior. The increase in incidents involving autocracies also shows that these attacks are not limited to democratic targets.

The most visible spike appears around 2017 and 2018, when incidents targeting democracies approach 100 in a single year. This period includes widely reported events such as election interference, espionage campaigns, and financial cyber operations. The spike is sharp and then declines somewhat, which likely reflects both real changes in activity and differences in how incidents are reported and attributed.

After 2021, the pattern shifts again. Incidents targeting autocracies increase quickly and exceed those targeting democracies by 2022. This change is closely tied to the cyber dimension of the war in Ukraine. More broadly, it suggests that cyber operations often expand alongside conventional conflict rather than replacing it. This is not something the regression models are designed to capture, but it stands out as an important pattern. It is also important to note that, based on the V-Dem Liberal Democracy Index, Ukraine has an average score of approximately 0.2 over this period and is therefore classified as an autocracy in this analysis, despite often being described as a democratic or hybrid regime in other contexts.

D. *Internet Penetration and Target Status*

The time series shows how patterns evolve, but it does not directly test the main claim that connectivity matters more than regime type. Figure 4 addresses this more directly by comparing internet penetration for all country-year values with no attacks and those with at least one attack. The figure uses a box plot with individual observations shown as scattered points.

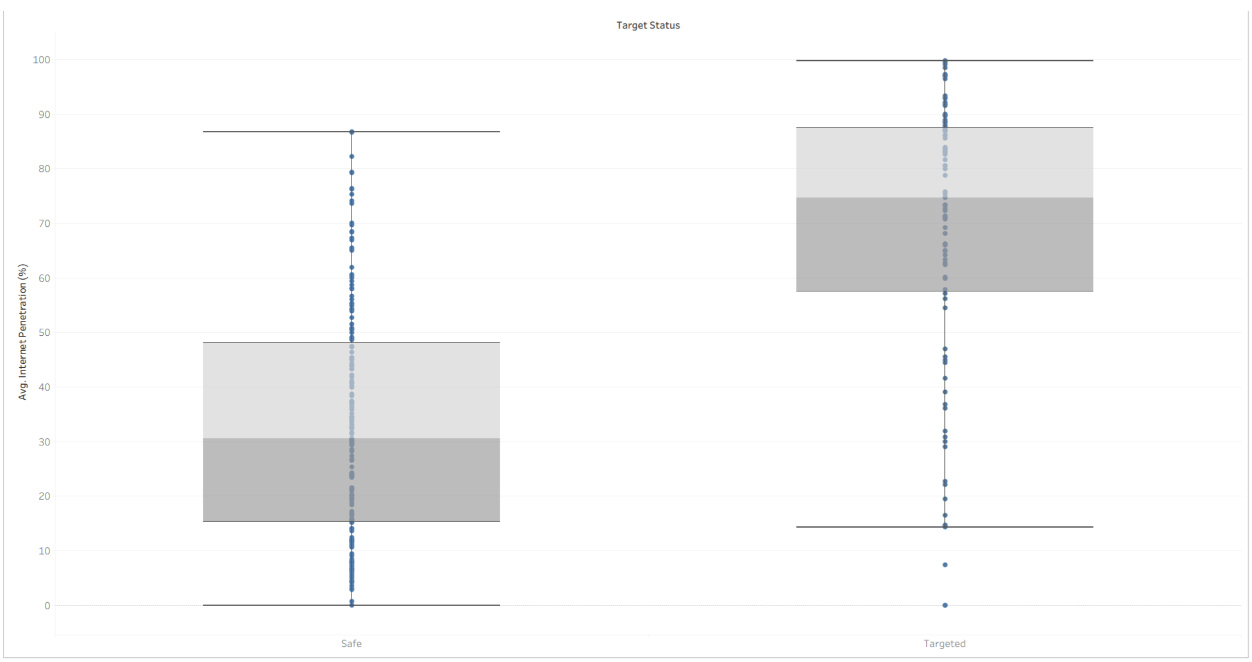

Fig. 4: Distribution of Average Internet Penetration (Jittered points show individual country-year observations. Safe indicates zero recorded incidents; Targeted indicates one or more recorded incidents.

This type of plot works well because the goal is to compare distributions between two groups. The individual points help show that the pattern is not driven by a few extreme values.

The difference between the groups is large. Country years with no attacks have a median internet penetration of about 30 percent, while those with at least one attack have a median closer to 74 percent. In practical terms, targeted countries tend to be much more connected.

The individual observations reinforce this. On the targeted side, most points are clustered at higher levels of connectivity. On the non targeted side, the spread is wider, reflecting the fact that many countries have zero incidents regardless of their level of internet use.

This figure provides a clear takeaway that higher connectivity is associated with a greater likelihood of being targeted. A regression analysis is needed to confirm whether this holds once other factors are included, but the basic relationship is already visible.

E. Internet Penetration, Attacks, GDP, and Regime Type

The previous figure focuses on connectivity alone. To see how it compares with other factors, Figure 5 combines internet penetration, attack frequency, GDP per capita, and regime type in a single chart. Internet penetration is shown on the horizontal axis, total incidents on the vertical axis, bubble size reflects GDP per capita, and color represents regime type.

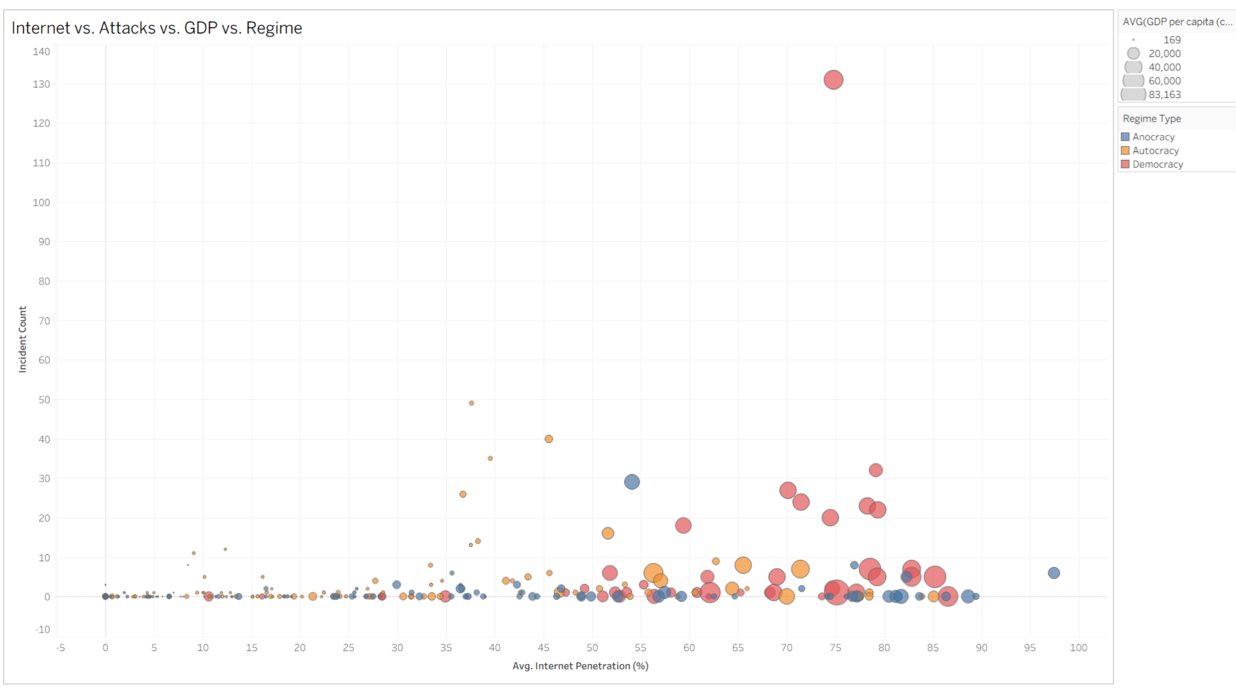

Fig. 5: Internet Penetration, Cyberattack Frequency, GDP per Capita, and Regime Type by Country. Bubble size represents average GDP per capita; color indicates regime type.

The overall pattern is consistent with earlier results. Most observations sit near zero incidents, and attack frequency increases as countries move to the right along the connectivity axis. There is no clear cluster of high attack countries at low levels of internet penetration.

The distribution of regime types is also revealing. If political systems were the main driver, we would expect to see democracies grouped at higher attack levels. Instead, both democracies and autocracies appear among the more frequently targeted countries, and they tend to share high connectivity. Some autocracies show high levels of targeting, while many democracies remain at zero when their connectivity is low. This weakens the idea that democracies are uniquely vulnerable.

GDP per capita shows a similar pattern. Wealth does not track closely with attack frequency once connectivity is taken into account. Some of the largest economies show relatively few incidents, while smaller economies appear at higher levels.

The comparison between the United States and Iceland helps illustrate this point. The United States combines high connectivity, wealth, and geopolitical importance, which corresponds to a high number of incidents. Iceland has very high internet penetration but far fewer incidents. This suggests that connectivity creates exposure, but other factors still shape targeting decisions. These visual patterns suggest that connectivity plays a central role, but they also highlight the need for a multivariate approach to separate overlapping factors.

F. Regression Results

The visualizations provide a useful starting point, but they do not tell us whether these patterns still hold once other factors are taken into account. Countries with high internet penetration also tend to be wealthier, more democratic, and located in regions with more geopolitical activity, so it is important to separate these effects. To do this, I used two OLS regression models. Table 1 reports the results.

TABLE I: Predictors of State-Sponsored Cyberattack Frequency

	Base OLS (1)	Full OLS (2)
Liberal Democracy Index (V-Dem)	-0.185	-0.451**
	(0.136)	(0.203)
Log GDP per Capita	0.018	-0.037
	(0.032)	(0.052)
Internet Penetration (%)	0.010***	0.010***
	(0.001)	(0.001)
Gov. Effectiveness (Detection Bias)	—	0.026
		(0.074)
Constant	-0.175	0.329
	(0.212)	(0.424)
Regional Fixed Effects	No	Yes
Observations	2,482	2,261
R^2	0.056	0.081
Adjusted R^2	0.055	0.076

Note: *p<0.1; **p<0.05; ***p<0.01

The most consistent result across both models is the effect of internet penetration. In Model 1, **a one percentage point increase in internet use is associated with about 0.01 additional incidents per country year. This relationship is significant at the 99% level**. This estimate does not change in Model 2 after adding controls for government effectiveness and region. That kind of stability suggests a strong and reliable link between how connected a country is and how often it is targeted.

The results for regime type are more surprising. In the base model, the coefficient on the V Dem Liberal Democracy Index is negative and is not statistically significant. This means that there is no clear support for the idea that democracies are targeted more often. However, once regional fixed effects are added in Model 2, the coefficient becomes larger in magnitude and statistically significant. Specifically, **A one-unit increase in regime type is associated with approximately 0.451 fewer cyber incidents per country-year, and this relationship is statistically significant at the 95% confidence level** once regional fixed effects are included. More democratic countries are associated with fewer incidents, not more. This suggests that earlier assumptions about democratic vulnerability may be driven by where democracies are located rather than by their political systems. Since democracies are concentrated in regions with high levels of cyber activity, it can look like regime type is the driver when the real factor is geography.

Economic development does not appear to play a major role. Log GDP per capita is not significant in either model, and the sign changes between specifications. This matches what we saw in the earlier visualizations, where wealth alone did not line up clearly with targeting frequency. Government effectiveness is also not significant. While this does not fully remove concerns about reporting bias in the data, it suggests that differences in detection and reporting are not driving the main results.

The R squared values are relatively low, ranging from about 5.6 percent to 8.1 percent, but this is not unusual for cross national work on cyber conflict. Many important factors, such as rivalries, alliances, and specific vulnerabilities, are difficult to measure at the country level. Even then, the models still point to a clear pattern. Internet penetration stands out as the strongest predictor, while the idea that democracies are inherently more vulnerable does not hold up in the data.

Taken together, these results do not support the original hypothesis. There is no consistent evidence that democracies are targeted more frequently than autocracies, and once regional factors are accounted for, the relationship actually moves in the opposite direction. This suggests that what initially appears to be a democracy effect is better explained by underlying geographic and structural patterns rather than regime type itself.

This result also raises some questions about how to interpret the relationship. One possibility is a kind of geopolitical shielding effect. Democracies, especially those tied to Western institutions like NATO, may benefit from intelligence sharing that reduces their overall exposure once regional context is taken into account. Another explanation is about how targets are chosen. Authoritarian states may actually be more attractive in some cases, since political and economic information is often more centralized and potentially easier to exploit. A third possibility is that the earlier pattern was driven by geography rather than regime type. Once regional differences are controlled for, that effect disappears and the underlying relationship becomes clearer. It is not possible to fully separate these explanations with the current data, but the results suggest that the idea of democratic vulnerability is less robust than it initially appears.

G. Marginal Effects

The regression table shows the direction and significance of the relationships, but it can be hard to interpret what they mean in practical terms. Marginal effects plots help make this clearer by showing how predicted outcomes change across the range of each variable.

Figure 6 shows predicted cyberattack frequency across different levels of the V Dem Liberal Democracy Index, with other variables held constant. The downward slope matches the regression results. As democracy increases, the predicted number of incidents decreases. A country with a very low democracy score has a predicted value of around 0.35 incidents per country year, while a highly democratic country is close to zero. The slight negative value at the top end is

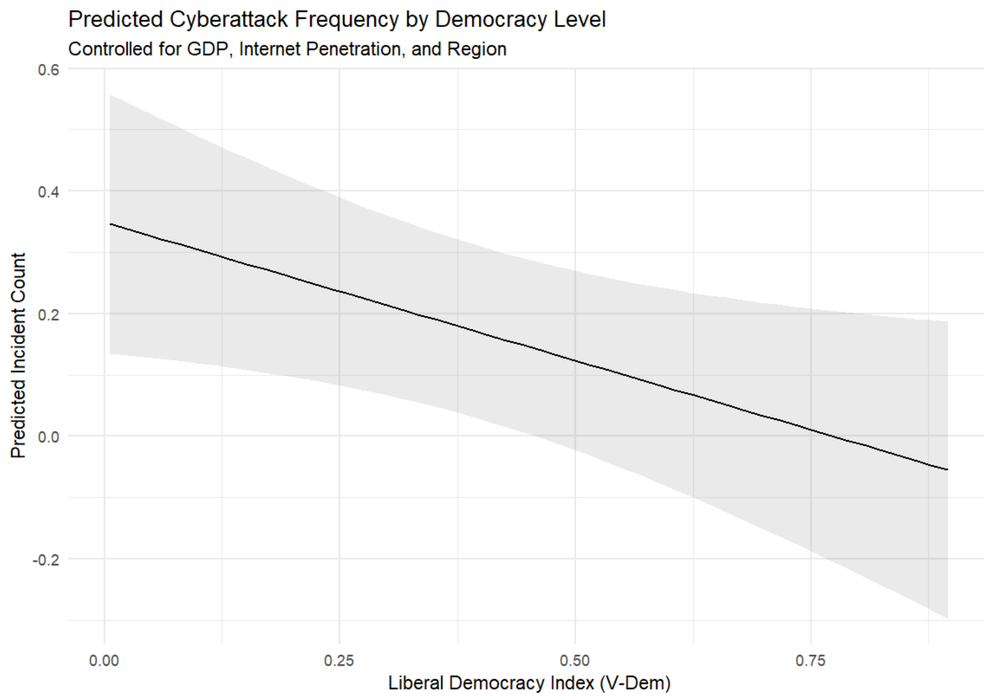

Fig. 6: Predicted Cyberattack Frequency by Democracy Level, controlling for GDP, Internet Penetration, and Region. Shaded area represents 95% confidence interval.

not meaningful in a real world sense. It reflects a limitation of using OLS with count data rather than an actual prediction of negative incidents.

The shaded area shows the 95 percent confidence interval. The interval widens at the extremes, where there are fewer observations, which means there is more uncertainty in those estimates. Even so, the overall pattern is clear. The relationship does not support the idea that democracies are more frequently targeted. In fact, it points in the opposite direction.

Figure 7 compares the coefficients from the base and full models. This makes it easier to see how the estimates change when regional controls are added.

The largest shift occurs for the democracy variable. In the base model, the estimate is close to zero and the confidence interval crosses zero, which indicates no clear effect. In the full model, the estimate moves to the left and the interval no longer overlaps with zero. This highlights how important the regional controls are. Without them, the model mixes together geography and regime type. Once region is accounted for, the relationship between democracy and targeting becomes clearer. Internet penetration shows almost no change between models, and the confidence intervals remain tight and far from zero. This reinforces the idea that connectivity is a strong and stable predictor. In contrast, GDP and government effectiveness both remain close to zero in both models, which supports the conclusion that they do not independently explain targeting frequency. Together, these plots reinforce the idea that connectivity, not regime type, is the most consistent factor shaping cyberattack targeting.

VI. Conclusion

This study examined whether regime type is a key predictor of how often countries are targeted by state sponsored cyberattacks. The common expectation, based on the democratic vulnerability

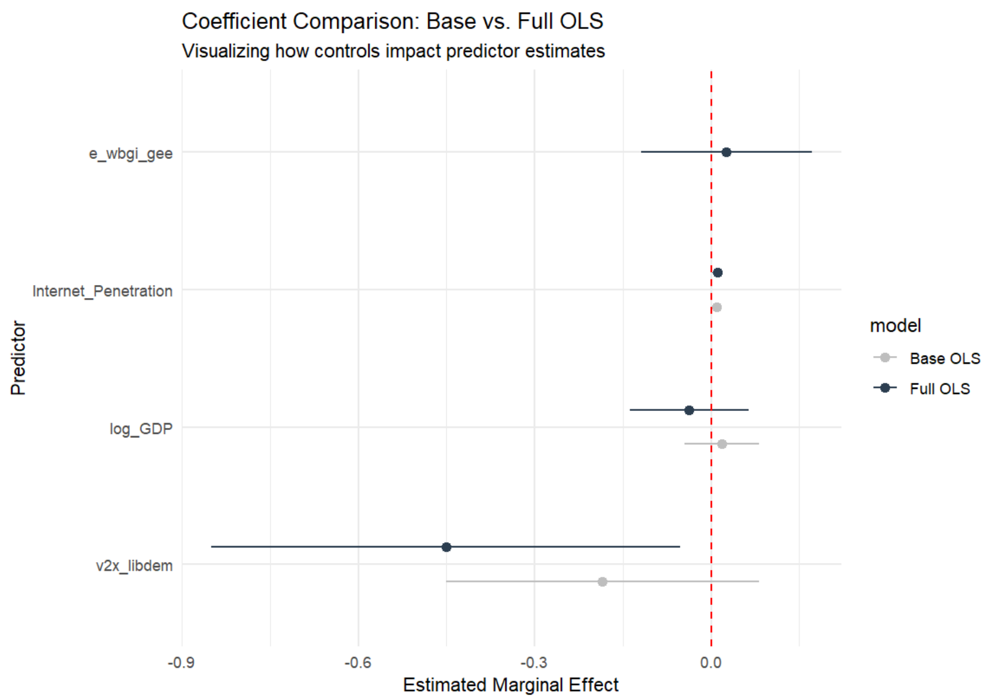

Fig. 7: Coefficient Comparison: Base OLS vs. Full OLS. Points show estimated marginal effects with 95% confidence intervals.

argument, is that open and wealthy democracies should be the most frequent targets. However, the results point in a different direction. Across nearly two decades of data, internet penetration stands out as the strongest and most consistent predictor of targeting. A one percentage point increase in internet use is associated with roughly 0.01 additional incidents per country-year ($p < 0.01$), and this relationship holds across all model specifications. In contrast, regime type is not significant in the baseline model and only becomes significant once regional fixed effects are added, at which point it is negatively signed. This suggests that once geographic clustering is taken into account, democracies are actually targeted less often than autocracies. Rather than simply failing to support the hypothesis, the results point in the opposite direction. At the same time, GDP per capita and government effectiveness are not significant in any specification, which suggests that wealth and institutional capacity do not explain much of the variation in targeting frequency. These findings suggest that exposure through connectivity matters more than political identity because attackers prioritize access to networks rather than regime characteristics.

One additional pattern that stands out is what appears to be a shift in targeting over time that lines up with major geopolitical events. In particular, there is a noticeable increase in attacks directed at autocratic states after 2021, which coincides with the escalation of the war in Ukraine. This suggests that cyber operations are often used alongside traditional military conflict rather than replacing it. It also points to a useful direction for future research, where conventional conflict could be explicitly modeled as a driver of cyber activity.

A. *Limitations and Future Work*

The main limitation of this study is the reporting bias built into the CFR Cyber Operations Tracker. Since the dataset relies heavily on Western open source intelligence, cyber incidents involving non Western countries are likely underreported. For example, operations occurring within regions like Central Asia or the Middle East may be less visible in the data than those involving Western targets. This is a limitation of the data rather than random measurement error. While regional fixed effects and the lack of significance for government effectiveness suggest that this bias does not fully drive the results, it cannot be completely ruled out. Future research would benefit from using alternative data sources, such as regional CERT reports or non Western intelligence collections, to test whether the same patterns hold under different reporting environments.

A second limitation relates to how cyber incidents are coded. The dataset treats all attacks equally, whether they target government ministries or private sector infrastructure like hospitals or firms. In reality, these types of targets may reflect different strategic logics, but they are not distinguished in the data. Future work could improve on this by separating incidents based on target type in order to better understand whether different factors drive attacks on public versus private systems.

The relatively low R squared values, ranging from 5.6 percent to 8.1 percent, also deserve attention. These models explain only a small share of the overall variation in targeting, which is not surprising given how many case specific factors shape cyber operations. Bilateral rivalries, alliance relationships, and specific technical vulnerabilities likely play a large role but are not captured in country level aggregates. The goal here is not to fully explain cyber conflict, but to identify structural patterns that have not been well-emphasized in prior studies.

Finally, another important limitation of this analysis is the absence of controls for interstate conflict and regional instability. As seen in Figure 3, the descriptive trends suggest that increases in state-sponsored cyber activity often coincide with periods of geopolitical tension and warfare, raising the possibility that conflict is an underlying driver of observed patterns. Future research should incorporate measures of conflict intensity or military disputes to better isolate the independent effect of regime type.

B. Final Remarks

The results are consistent across all specifications. Internet penetration is the strongest and most reliable predictor of cyberattack targeting, while regime type does not behave as the democratic vulnerability thesis would predict. Once regional differences are accounted for, democracies are not more frequently targeted and may actually be targeted less often. For policymakers, this implies that cyber risk is driven more by connectivity than by political system. In practical terms, expanding digital infrastructure appears to increase exposure, which suggests that cybersecurity policy should focus on strengthening technical defenses and network resilience rather than relying on regime based assumptions about vulnerability.

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VII. Word Count

Word Count: 6383